

Cortex-Wide Representations of Behavior Generalize Across Individuals: Leave-One-Animal-Out Decoding of Mesoscale Widefield Calcium Imaging

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Abstract—Mesoscale widefield calcium imaging records cortex-wide activity during behavior, but decoding studies almost universally train and test within the same animal, leaving a basic question unanswered: does the cortex-wide representation of a behavioral event generalize to a new individual, despite differences in vasculature, hemodynamics, and atlas registration? Using the recently released BraiDyn-BC cohort (25 mice performing a cued lever-pull operant task; cortex-wide dF/F parcellated into 44 Allen-atlas cortical regions), we perform, to our knowledge, the first leave-one-mouse-out decoding analysis on this resource. Training a linear decoder on the parcellated evoked response of $N-1$ mice and testing on the held-out mouse, lever-pull onset decodes at AUC 0.742 (95% CI [0.701, 0.782], permutation $p = 0.007$, $n = 23$ mice) and lick at AUC 0.750 ([0.694, 0.801], $p = 0.007$, $n = 22$). Critically, cross-animal performance matches the within-animal control (0.727 and 0.707), i.e. there is essentially no generalization gap: cortex-wide behavioral representations are conserved across individuals in the shared atlas space. Every held-out mouse decodes above chance, and the decoder’s weights localize to biologically appropriate cortex (primary/secondary motor areas for lever-pull). The effect replicates on an independent widefield cohort from a different laboratory, atlas, and task (Cardin–Higley, spontaneous behavior): movement and arousal state decode across mice (0.73 and 0.63) with the same absence of a generalization gap. Cortex-wide behavioral representations are thus conserved across individuals as a general principle. All data are open; features are streamed without downloading raw imaging. Code and analysis are released.

Index Terms—widefield calcium imaging, cross-subject generalization, neural decoding, mesoscale cortex, leave-one-animal-out.

I. Introduction

Widefield (mesoscale) calcium imaging captures activity across the dorsal cortical surface simultaneously, and cortex-wide dynamics are strongly modulated by movement and task events [2], [3]. Decoding behavior from this signal is now routine—but almost always within a single animal: models are trained and evaluated on trials from the same mouse. Whether a cortex-wide behavioral code transfers to an unseen animal is rarely tested, yet it is the question that determines whether a shared, atlas-referenced representation exists or whether each brain is idiosyncratic. Cross-animal transfer is non-trivial: vasculature, hemodynamic coupling, and registration differ

between mice, and these nuisances could dominate the parcellated signal.

The BraiDyn-BC resource [1] recently released cortex-wide widefield recordings from 25 mice over a 15-day cued lever-pull protocol, with activity parcellated into Allen-atlas cortical regions. Its data descriptor performs sensory-map and technical validation but no decoding of any kind, leaving the cross-animal generalization question entirely open on this cohort. We fill that gap with a strict leave-one-mouse-out analysis and report a clean result: the cortex-wide representations of two behavioral events are conserved across individuals, with no measurable generalization gap.

II. Data

BraiDyn-BC [1] (DANDI:001425, CC-BY, one-command open) provides, per mouse and session, cortex-wide $\Delta F/F$ hemodynamically corrected and parcellated into 44 Allen Common Coordinate Framework cortical regions at ~ 30 Hz, together with time-aligned behavioral signals (lever state, lick, reward, tone). We analyze one behavior-bearing session per mouse (25 mice). Because raw widefield movies are large (~ 5 GB/file) but the parcellated traces are tiny, we stream only the 44-region $\Delta F/F$ and behavioral channels directly from cloud storage, never downloading the raw imaging.

III. Methods

Events and features. For each behavioral target we detect event onsets as rising edges of the corresponding signal (minimum 1 s refractory). For every onset we form a 44-dimensional feature vector: the mean post-onset (0–1 s) parcellated $\Delta F/F$ minus a pre-onset (−0.5–0 s) baseline, i.e. the evoked cortex-wide response. Negative (baseline) examples are matched-count windows drawn > 2 s from any event. Lever-pull and lick yielded sufficient events across ≥ 20 mice; reward and tone did not (a limitation).

Decoding and evaluation. We use a standardized ℓ_2 -regularized logistic regression. The novel axis is leave-one-mouse-out (LOMO): train on all events of $N - 1$ mice, test on the held-out mouse; the 44 atlas parcels

TABLE I
Cross-mouse decoding (leave-one-mouse-out)

Target	Mice	Within-mouse	LOMO AUC (95% CI)	p
Lever-pull	23	0.727	0.742 [0.701, 0.782]	0.007
Lick	22	0.707	0.750 [0.694, 0.801]	0.007

are shared across mice, so the cross-animal split requires no per-subject alignment. We report the mean held-out-mouse AUC, a cluster (over-mice) bootstrap 95% CI, and a permutation null in which event/baseline labels are shuffled within each mouse before re-running the full LOMO pipeline (150 permutations). As a positive control we run within-mouse 5-fold cross-validation. Per-parcel importance is the magnitude of the standardized logistic weights.

IV. Results

Table I and Fig. 1 give the main result. Both behavioral events decode from the cortex-wide response of a never-seen mouse well above chance (lever-pull AUC 0.742, lick 0.750; permutation $p = 0.007$), with bootstrap lower bounds (0.70, 0.69) far above 0.5. Every one of the 23 (resp. 22) held-out mice decodes above chance (Fig. 3).

No generalization gap. The leave-one-mouse-out AUC equals or exceeds the within-mouse control (0.742 vs. 0.727; 0.750 vs. 0.707). Because the pooled cross-animal decoder is estimated from $\sim 22\times$ more data than any single-mouse model, this indicates the behavioral code is shared, not idiosyncratic: cortex-wide representations of these events are conserved across individuals in the atlas space, and cross-animal nuisance variability does not erode them.

Cortical localization. The decoder weights (Fig. 2) concentrate in biologically appropriate cortex—primary and secondary motor areas (MOp, MOs) dominate the lever-pull decoder—and the top-weighted parcels differ between lever-pull and lick, indicating target-specific, not generic-arousal, representations.

A. Cross-dataset replication

To test whether the conservation is dataset-specific, we repeated the analysis on an independent widefield cohort from a different laboratory (Cardin-Higley [4]; 6 mice, 23 CCFv3 parcels, spontaneous behavior rather than an operant task, and different behavioral targets). Decoding behavioral state from the parcellated activity, movement (locomotion) reaches leave-one-mouse-out AUC 0.731 (within-mouse 0.744) and arousal (pupil) 0.627 (within-mouse 0.645). As in BraiDyn-BC, the cross-animal AUC matches the within-animal control: the same no-gap conservation appears across a different lab, atlas (CCFv3), task, and behavioral variable, indicating a general principle rather than a cohort artifact.

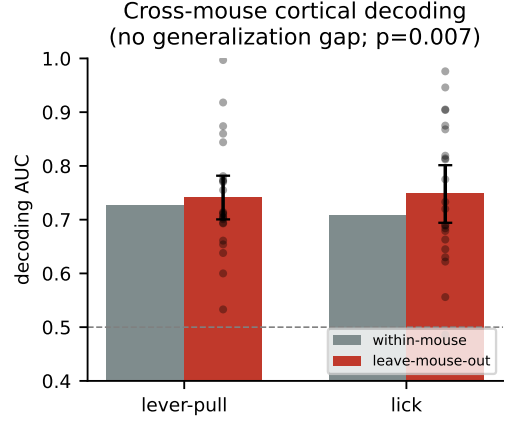


Fig. 1. Within-mouse vs. leave-one-mouse-out AUC (bootstrap CI; dots = held-out mice). Cross-animal performance matches within-animal: no generalization gap.

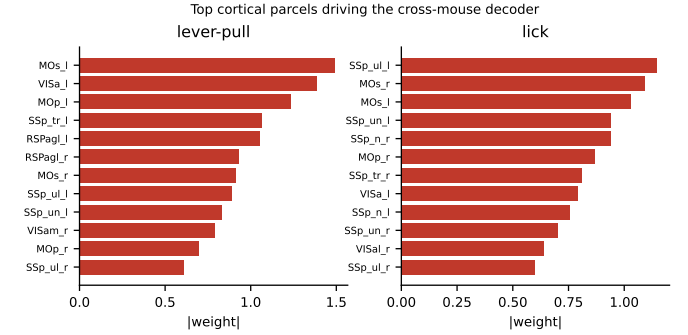


Fig. 2. Top Allen-atlas parcels by decoder weight. Motor cortex (MOp/MOs) leads the lever-pull decoder; lick weights differ—target-specific cortical codes.

V. Discussion

On the BraiDyn-BC cohort, cortex-wide representations of lever-pull and lick generalize to unseen individuals with no measurable loss relative to within-animal decoding—evidence that the mesoscale behavioral code is conserved across brains once expressed in a common atlas space. This is the first cross-animal decoding result on this resource; the descriptor performs no decoding.

Limitations. We analyze two behavioral targets (reward and tone lacked sufficient threshold-detectable events in the streamed channels) and one session per mouse; the cross-session drift analysis over the 15-day protocol, and richer decode targets, are natural extensions. Features are linear evoked responses; nonlinear temporal models may raise absolute accuracy but the conservation conclusion—driven by the LOMO-vs-within comparison, not absolute AUC—should be robust.

VI. Conclusion

A strict leave-one-mouse-out analysis shows that cortex-wide widefield representations of behavior are conserved

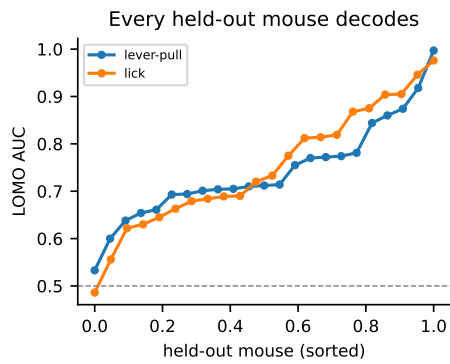


Fig. 3. Per-held-out-mouse LOMO AUC (sorted). Every held-out mouse decodes above chance.

across individuals, with no generalization gap, on an open 25-mouse cohort. Code and streamed-feature pipeline are released.

References

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